EQUIPMENT REPAIR HISTORY TRACKER

Team research

Customer research behind the client brief, by team member

2 November 2026

1 **Shreyaa** — 6 sources
How it's done today, where it goes wrong, and what it costs
Current process, failure points and costs; which tool rule catches which problem.

2 **Harshitha Reddy Gundavarapu** — 11 sources
Rules that apply: contract, data handling, safety and software updates
The obligations behind visit records and how each one shaped the build and release.

3 **Natalie King** — 15 sources
Who the users are and the words they use
Roles, field service vocabulary, part numbers and validation rules.

4 **Julian Marchington** — 15 sources
How records are kept
Why a full history is needed, how long to keep it, and who may see or change it.

How to read this document

Each member's research is a separate part, in their own words. Numbers in square brackets, such as [3], point to the numbered References at the end of that member's part, and each number is a link. Each part has its own numbering, so [3] in one part is not the same source as [3] in another.

Changes made when compiling: four sources in Shreyaa's and Harshitha's parts were named in the original document without a web address (Ventavid callback rate, AccountingTools, ISMS.online controls 8.32 and 8.13). Their addresses were added and are marked in the References. One sentence addressed to Shreyaa as an instruction ("Write this up as your brief section...") was removed. Nothing else was cut from their parts. Notes such as "Add page and quote after reading the page" are kept, because they mark checks still to do.

Natalie's and Julian's parts are shortened versions of their full reports, each limited to 15 sources. They keep the main findings, the draft brief sections, assumptions log rows, client questions, openers and model answers. The full reports, with every source, are in the project files under natalie-research/reports and julian-research/reports.

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Shreyaa: How it’s done today, where it goes wrong, and what it costs

Researched and written by **Shreyaa**

The industry measures this problem with two numbers, first-time fix rate and repeat (callback) visits, and the main causes are missing parts and history nobody can see. Every rule you built maps to one of those causes. I opened each page below and checked the claims against it.

How it’s done today

- Technicians fill in a visit record (machine, fault, parts used, outcome). In our client’s case, that ends up typed into a spreadsheet. The stockroom keeps a separate count.
- Service teams judge themselves on **first-time fix rate (FTFR)**: the share of jobs finished on the first visit with no follow-up. IBM puts the typical figure at about 80%, so roughly one job in five needs another trip. (IBM [1])
- The flip side is the **repeat visit** or **callback rate**: return trips to a job the first visit didn’t close, divided by total jobs, counted inside a set window. There is no single standard window. Thirty days is common in residential service; ninety days is common for capital equipment. The advice is to pick one, write it down and stick with it. (Ventavid: callback rate [2], Ventavid: repeat visit [3])
- Stock is managed with a **reorder point**: the stock level that triggers a new order so you don’t run out, while keeping inventory low. (AccountingTools [4])

Where it goes wrong with spreadsheets

- **Nobody lines up the history.** A repeat failure only shows if someone sorts every visit by serial number and counts the dates. With visits scattered across sheets, a switch can fail three times in a month and nobody connects them.
- **Spreadsheets carry errors.** Panko’s review of spreadsheet audits found that the more careful recent audits turned up errors in at least 86% of the spreadsheets they checked, with cell error rates of roughly 1% to 2%. (Panko, 2000 [5]) One mistyped quantity silently corrupts a stock count.
- **The stockroom finds out too late.** If the count says spares exist but they were used, the technician discovers it on site.

What it costs

- Repeat visits raise operating costs and tie up people who could be on other jobs. Missing parts are named as a direct cause of failing to fix on the first visit. (IBM [1])
- PTC cites Aberdeen Group benchmarks: best-in-class teams fix on the first visit 88% of the time, average teams 80%, laggards 63%. PTC lists lack of spare parts as one of three main reasons a first fix fails. (PTC [6])

- PTC also says a failed first fix takes an extra 1.6 dispatches at $200 to $300 per truck roll. **Be careful with this one:** PTC gives no source for those two numbers. Use them as "one vendor estimate" only, or leave them out.
- Our client hasn't given us their own costs, so we don't invent a dollar figure. We talk about work removed: repeat call-outs, recounting, digging through sheets.

Which rule catches which problem

Rule	Real problem it catches	Research behind it
Repeat failure	A fix that isn't holding, so the client keeps paying for visits	Repeat visit / callback rate, measured in a fixed window (Ventavid), lowers FTFR (IBM)
Low stock	The shelf is about to run out before the next job	Reorder point (AccountingTools); missing parts hurt first-time fix (IBM, PTC)
Stock shortfall	The records say we used parts we didn't have, so the count is wrong or parts went untracked	Spreadsheet error rates (Panko)
Unknown part (1.1.0)	A technician logged a part that isn't on the stock list, so its usage is invisible to the stockroom	Same data-quality risk (Panko); stock you can't see can't trigger a reorder

How the research shaped what we built

1. **The window is a setting, not code.** Because sources disagree on 30 versus 90 days, repeat_failure_count and repeat_failure_window_days live in settings.json. We started at 3 in 30, matching the brief's "three times in a month."
2. **Low stock fires at the reorder level, not at zero.** That's what a reorder point is for: warn while there's still time to order.
3. **Shortfall lets the count go negative** instead of blocking the save, so the gap stays visible for someone to investigate.

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Harshitha: Rules that apply: contract, data handling, safety and software updates

Researched and written by **Harshitha Reddy Gundavarapu**

This section covers the contract, data handling, safety and software-update rules that apply to a service company's visit records and customer site information, and what each one changed in how the tool was built and released.

The tool flags and reports only. It never orders parts, schedules a repair, or decides that equipment should be replaced. The service manager decides.

1. Contract: records of work done

The rule. A service company must be able to show what work it did on which equipment. Contracts commonly require accurate and complete service records, kept under applicable law and the customer's written retention policy. Sample clause: Law Insider, record maintenance and availability [1]. This is a sample clause, not law, and it names no retention period.

What it changed. Every visit and the parts used are stored in SQLite and are never overwritten, so earlier service records can always be retrieved by serial number (service-tracker history SERIAL).

2. Data handling

The rule. Visit records hold equipment serial numbers, customer site information, technician details, repair history and parts used. The main duties are confidentiality, integrity, availability and retention.

GDPR Article 5 is the clearest published statement of these principles. Personal data must be "adequate, relevant and limited to what is necessary", kept "for no longer than is necessary", and processed securely, "including protection against … accidental loss, destruction or damage" (GDPR Article 5 [2]). The European Commission adds that data "must be stored for the shortest time possible" and that organisations should set time limits to erase or review it (European Commission, principles of GDPR [3]).

The law that actually applies depends on the customer, the information and the country, so these are safeguards and not proof of compliance.

What it changed.

- Confidentiality: synthetic data only, only the columns in the shared contract, and the Docker image runs as a non-root user.
- Integrity: records are validated before saving, bad rows are rejected with a reason, and each visit is saved in one transaction so there are no half-saved visits.
- Availability: the database is on a Docker volume, and the upgrade procedure includes a backup checked with PRAGMA integrity_check.
- Retention: the tool has no delete or expiry feature. How long to keep records is the customer's decision.

- Access control: the tool has no login. Access depends on who can reach the files and the volume. The client gave no authentication requirement.

3. Safety

The rule. Technicians repairing servers and network equipment face electrical hazards and must follow workplace safety rules and customer-site procedures. OSHA 29 CFR 1910.333 says live parts “shall be deenergized before the employee works on or near them”, unless that is infeasible or adds hazards, and then “other safety-related work practices shall be used” (OSHA 1910.333 [4]). OSHA 29 CFR 1910.147 covers the control of hazardous energy during servicing and maintenance, and whether it applies depends on the equipment and the work (OSHA 1910.147 [5]). *[Add page and quote after reading the page.]*

What it changed. Nothing about physical safety, because the tool does not enforce safety procedures. It keeps reliable repair history and flags repeat failures, low stock, stock shortfalls and unknown parts, so managers can investigate problems early. Technicians must still follow safety procedures.

4. Software updates

The rule. The company depends on the repair history, so an update must not lose it, must be tested, and must be reversible.

- NIST SP 800-53 CM-3 describes change control as “the systematic proposal, justification, implementation, testing, review, and disposition of changes” and says the organisation “retains records of configuration-controlled changes” (NIST SP 800-53 CM-3 [6]).
- A summary of ISO 27001 control 8.32 lists testing, “a fall-back procedure” and records of all changes (ISMS.online, control 8.32 [7]).
- A summary of control 8.13 says backup jobs should be tested regularly and kept “physically separate from the source data” (ISMS.online, control 8.13 [8]).
- Docker says a volume’s contents “exist outside the lifecycle of a given container” (Docker volumes [9]).
- SQLite says its backup API gives “a consistent and up-to-date snapshot of the original”, while a plain file copy can leave a corrupted backup after a power or system failure (SQLite Online Backup API [10]).
- The NIST Cybersecurity Framework 2.0 gives general guidance on protecting information and recovering from incidents (NIST CSWP 29 [11]). *[Add page and quote after reading the page.]*

What it changed.

- The database lives on a named Docker volume, so history survives when a container is removed.
- GitHub Actions runs the tests and builds the image on every pull request.
- Releases 1.0.0 and 1.1.0 each have a pinned version, a git tag and a CHANGELOG.md entry.
- The upgrade procedure takes a backup with Connection.backup and checks it with PRAGMA integrity_check.

- Release 1.1.0 adds one rule and changes no tables, so 1.0.0 can still read the history. We ran the upgrade and rollback drill from a clean volume twice. If a future release changes tables, the backup would be restored instead.

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Natalie: Who the users are and the words they use

Researched and written by **Natalie King**

Technicians write the records, the manager reads them, and the words do not yet match

In the client's firm, **ten technicians create the visit records, and the service manager and director read them**. The research shows the same pattern across field service: the engineer on site fills in the job record, an office or manager role schedules and reports, and a stock role reorders parts. In a firm of twelve people there is no separate dispatcher or storeman, so the service manager almost certainly does dispatch, stock control and reporting as one job. Technicians need the device's history and the right parts before they arrive. The service manager needs repeat faults, first-time fix and low-stock warnings, all of which depend on complete records. Our tool's columns hold the core of a real job sheet: a record ID, the device, the date, the fault as a code, the parts and quantities, and an outcome. The vocabulary is where the clearest gaps are. **UK sources say "field service engineer", "job sheet" and "stock on your vehicle"**, while the client says "technician" and "paper job sheets". The outcome `failed` clashes with the international definition of "failure", and there is no "no fault found" or "awaiting parts" value. No source supports the exact part-number format "P-" plus four digits, but neutral sources support a fixed internal format checked by a full-match pattern.

How to read this report. The research was done on 2 November 2026 through a web-fetch tool that returns extracts of each page, so no quote has been checked character by character against the original. Where only an abstract was seen, the report says so. Vendor sources are labelled and used only to show common practice. No source studied IT field engineers (servers, switches, routers, firewalls) specifically; most evidence comes from general maintenance and field service.

Four roles use the records, and in a small firm one person holds three of them

Microsoft's documentation for its Dynamics 365 Field Service product (vendor) gives a ready-made set of roles. The frontline worker role is "Designed for frontline workers who carry out work orders for customers onsite". The dispatcher role is for "users who are responsible for scheduling jobs and assigning work orders to frontline workers". The administrator role is "Intended for IT administrators and service managers who configure the system for the rest of the organization". The inventory role is for "inventory managers who oversee inventory on service vehicles, purchase and reorder inventory" (Microsoft Learn [1], vendor).

Small firms combine these roles. A UK job advert on the government's Find a Job site asks a field service engineer to "Maintain adequate stock on your vehicle", "Fulfil repairs with stock from vehicle", and "Maintain accurate paperwork via laptop including time sheets, job sheets, stock paperwork" (DWP Find a Job, 2025 [2]; employer advert, not IT). So the technician is also their own van stock controller. The client named only a service manager and a director, so **our inference is that the service manager is dispatcher, stock controller and report reader in one.** That should be confirmed with the client.

Role	Creates or reads	Needs from the records	Client's equivalent
Field technician / engineer	Creates visit records; uses parts	The device's past visits and faults before arriving; the right part in the van	10 technicians
Dispatcher / office	Creates the job; reads status	Who is going where, and what is outstanding	No named role; probably the service manager
Service manager	Reads everything; configures	Repeat faults, first-time fix, KPIs. NIST authors found "data quality often impacts an analyst's ability to calculate important Key Performance Indicators (KPIs)" (Conte et al., 2021 [3])	Service manager
Stock / inventory manager	Reads parts used; updates stock; reorders	What was used, what is low, what is on the vans	Probably the service manager, with technicians handling van stock
Director / owner	Reads reports	Cost, profit, contract performance. **No source found**	Director

Our tool serves two of these needs directly: the repeat-failure flag for faults that keep coming back on one device, and the low-stock flag for the stock role. Both depend on the fields that peer-reviewed research finds are most often missing or wrong: equipment, fault and action (Hodkiewicz & Ho, 2016 [4], abstract only).

UK engineers say "job sheet" and "van stock", and international standards separate fault from failure

UK sources lean towards "engineer". A UK vendor says "Some people call it a job sheet, others say works order or service sheet" (Fieldmotion [5], vendor). The UK job advert uses "Field Service Engineer", "job sheets" and "stock paperwork" (DWP Find a Job [2]). The client's "technician" is also normal UK usage.

For faults, the most authoritative free source is the International Electrotechnical Commission's vocabulary, Electropedia (IEC 60050-192). A **failure** is "loss of ability to perform as required", and "an event that results in a fault" (IEV 192-03-01 [6]). A **fault** is "inability to perform as required, due to an internal state" (IEV 192-04-01 [7]). In plain words: the failure is what happened, and the fault is the broken state the technician finds.

How technicians actually write is less tidy. NIST researchers describe maintenance free text as "often unstructured, filled with technical content, jargon, mispellings [sic], and abbreviations" (Sexton & Brundage, 2019 [8]). **No source recorded the words IT field engineers use for servers, switches and their parts.** That is the biggest gap in this research. The team's own sample stock list already uses part words such as "PSU 500W" and "DIMM 32GB", while the problem codes use different words for the same things (PWR, MEM), so a technician who thinks "PSU" has to translate it to "PWR".

Term	Plain meaning	How our tool uses it / mismatch
Failure	The event of losing the ability to work (IEC)	**Clash:** outcome `failed` describes the visit, not the equipment
Fault	The broken state found on site (IEC)	Closest to `problem_code`
Job sheet / works order / service sheet	The record of one job (Fieldmotion, DWP advert)	Client's "paper job sheets"; the tool calls it a visit

Term	Plain meaning	How our tool uses it / mismatch
Field service engineer / technician	The person on site	Tool column `technician`, matching the client's word
Serial number / asset tag	The maker's identifier vs the owner's own tag	Tool's `serial_number`. Sample values (SW-4411, SRV-2040) look like asset tags. **Ask the client**
No fault found (NFF)	A part was removed but the fault could not be traced (Khan et al.)	**Missing outcome**
First-time fix	A job finished on the first visit (PTC)	Can be roughly estimated from serial, date and outcome

A fixed internal part number is good practice, but "P-" plus four digits has no source

The strongest neutral example of a fixed-format internal number is NATO's: "Each Item of Supply is identified in the NCS by a unique 13 digit NATO Stock Number (NSN)" (NATO NCS Fact Sheet [9]), which the fact sheet ties to preventing duplicate entries. **No source recommends "P-" plus exactly four digits.** It is a project convention taken from the sample stock list, and the client should confirm it.

The case for checking the format comes from secure-coding guidance. OWASP says "Define what the application accepts and reject values outside those rules", and "Use regular expressions for simple, structured fields", requiring "a match of the entire value" (OWASP Input Validation Cheat Sheet [10]). Our loader does both halves. The pattern `^P-\d{4}$` from `settings.json` is the syntax check, and the release 1.1.0 "unknown part" flag (a correctly formatted part not in the stock list) is the meaning check. Spaces are stripped and letters uppercased first, so "p-1180" is read as "P-1180".

Old and manufacturer numbers are a real problem. "Supersession is the term given to the replacement of one part number by a newer number", and parts systems such as Volvo's show the replacement when an old number is entered (Volvo Fusion manual [11], software publisher). Our tool has no cross-reference table. A manufacturer number in the wrong shape is rejected; an old internal number in the right shape is stored and flagged as unknown. A table mapping old and manufacturer numbers to P-nnnn is the obvious next step.

The outcome list mixes results and actions

The tool's four outcomes are `fixed`, `failed`, `replaced_part` and `escalated`. **`failed` is ambiguous**: in IEC 60050-192, failure is something the equipment does, so `outcome=failed` could be read as "the equipment failed" rather than "the visit did not fix it". **`replaced_part` is an action, not a result**, and overlaps with `fixed`. Two outcomes technicians are likely to need are missing. The first is **no fault found**: "Units tagged as 'NFF' are evidence that a serviceable component was removed" (Khan et al., 2014 [12], abstract only). The second is **awaiting parts**: PTC defines first-time fix as fixing the issue without needing "additional expertise, information, or parts", and names "Lack of spare parts" as a common cause of failing it (PTC [13], vendor). Without these values, technicians will log both cases as `failed`, and the manager cannot tell a hard fault from a missing part.

`problem_code` also mixes three kinds of thing: components (DISK, FAN, MEM, NIC, PORT, PWR), a symptom (OVERHEAT) and a software area (FIRMWARE), and any value is accepted, so a typo such as "FANS" would be stored as a new code. The practical fix is to define the code as "the main component or area at fault" and validate it against the eight known values.

Every current rejection rule is a completeness or validity check

The UK Government Data Quality Framework defines completeness (“the degree to which records are present”) and validity (“the degree to which the data is in the range and format expected”), and recommends improving quality “at source through validation rules”, solving issues “at source, rather than apply temporary fixes” (UK Government Data Quality Framework, 2020 [14]). On dates, the W3C says “The format MM/DD/YY is unique to the United States”, “Most of Europe uses DD/MM/YY”, and ISO 8601’s YYYY-MM-DD “is unambiguous” (W3C [15]). A UK paper sheet dated 03/04/2026 could be March or April, so the loader should not guess.

Loader rule (rejects the row, reports line number and reason)	Dimension
`visit_id`, `serial_number` or `part_number` missing	Completeness
`visit_date` not exactly YYYY-MM-DD, or not a real date (e.g. 2026-02-30)	Validity (format and range)
`quantity_used` not a whole number, negative or zero	Validity (type and range)
`outcome` not fixed / failed / replaced_part / escalated	Validity (allowed values)
`part_number` not matching `^P-\d{4}$`	Validity (format)
A column missing from the file	Completeness (whole file; no row is loaded)
Stock file: same `part_number` twice	Uniqueness

Not yet checked: rows sharing a `visit_id` that disagree on serial, date or outcome; placeholder serials such as “N/A”; and dates in the future. Because every row needs a part, a visit with no parts (an inspection or firmware update) cannot be recorded.

Conclusion

The client described two readers, but the records really have one writer group under time pressure and one overloaded manager who is dispatcher, stock controller and analyst at once. That makes the quality of what technicians type the deciding factor, and the peer-reviewed work-order studies show it is the weakest link. Checks at intake are therefore the only place the manager’s reports can be protected. The cheapest, highest-value changes are to define each code in plain words, add “no fault found” and “awaiting parts”, validate `problem_code`, and check that rows sharing a visit agree. The single most important open question is one only the client can answer: what words do their technicians actually write on the paper sheet today.

Draft client brief section: Who the users are and the words they use

Researched and written by Natalie King

Who uses the records. Your ten technicians create the visit records, and your service manager and director read them. This matches field service practice, where the engineer on site completes the job record, the office schedules it, and an inventory role reorders parts [1]. In a firm your size, one person usually covers dispatch, stock and reporting; we assume this is your service manager. Technicians often keep their own van stock and “stock paperwork” [2].

What each user needs. Technicians need a device’s past faults before they arrive. The service manager needs repeat faults and low-stock warnings. Both depend on complete records: peer-reviewed studies find real work orders often have “missing or incorrect functional location,

failure mode, maintenance action and WO status fields” [4], and that poor data “often impacts an analyst’s ability to calculate important Key Performance Indicators” [3].

The words. UK sources say “field service engineer”, “job sheets” and “stock on your vehicle” [2]; your word “technician” is also common. International standards define a failure as “loss of ability to perform as required” and a fault as the resulting “inability to perform” [6, 7]. We use “fault” for what a technician finds on site.

Part numbers. We assume your internal format is P- plus four digits. No standard sets this format, but fixed formats checked by a full-match pattern are recommended practice [10], and NATO uses a fixed-length stock number to prevent duplicates [9]. Makers replace part numbers over time (“supersession”) [11].

What this means for the tool. The tool rejects a row, with its line number and reason, if the visit ID, serial number or part number is missing, the date is not YYYY-MM-DD, the quantity is not a positive whole number, the outcome is not fixed, failed, replaced_part or escalated, or the part number is not P-nnnn. These are completeness and validity checks in the UK Government Data Quality Framework [14]. Not yet built: a “no fault found” outcome, which is a recognised maintenance category [12]; a check that rows for one visit agree; and a table linking old part numbers to current ones.

The numbers in this section point to the References at the end of Natalie’s research.

Assumptions log rows

Decision	What we assumed	Source or reason	Who decided
What makes a record bad	A row is rejected, with line number and reason, if a required field is missing, the date is not a real YYYY-MM-DD date, the quantity is not a positive whole number, the outcome is outside the allowed set, or the part number does not match `^P-\d{4}$`. A file missing a column is refused whole.	The brief gives missing serial, negative quantity and invalid date. The others follow the UK Government Data Quality Framework and OWASP allow-list validation. **Gaps:** no cross-row consistency check; “N/A” serials accepted; `problem_code` not validated.	Natalie
Part number format is P- plus 4 digits	Every valid internal part number is “P-” followed by exactly four digits, after stripping spaces and uppercasing.	Taken from the sample stock list; pattern held in `settings.json`. Supported in principle by OWASP and NATO’s fixed-length NSN. **No source recommends this exact format.** Future work: a cross-reference table for old and manufacturer numbers.	Natalie
Outcome values fixed / failed / replaced_part / escalated	These four cover every visit result.	Client brief and team design. **Found weak:** `failed` clashes with the IEC meaning of failure; `replaced_part` is an action; “no fault found” (Khan et al.) and “awaiting parts” (PTC) are missing. Future work: define each value, add the two missing ones.	Natalie
Who the users are	10 technicians create visit records; the service manager and director read history. The service manager probably also does dispatch and stock control.	Client meeting; Microsoft’s field service roles (vendor) and the UK job advert. Combined roles are our inference. **No source found for the director’s needs.**	Natalie

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Questions for the client

1. On your paper job sheets, what identifies the equipment: the maker's serial number (or Dell "service tag"), or your own asset tag? Our sample values such as SRV-2040 look like asset tags.
2. What does a technician write on the sheet today? Could you share one or two blank or anonymised sheets?
3. When a visit ends without a fix, what do technicians write? Do they distinguish "no fault found", "waiting for a part" and "needs a return visit"?
4. Are all your internal part numbers in the P- plus four digits format? Do technicians sometimes write the maker's number instead? Should the tool map those to your number, or reject them?
5. Who keeps the stock list today, and do technicians carry van stock? Should van stock count towards "quantity on hand"?
6. How often does a visit use no parts at all, for example an inspection or a firmware update? Should those visits be recorded?

One-minute presentation opener

> "Good morning. Today we're showing a tool that turns your paper job sheets into a searchable repair history. It loads visit and stock files, flags devices that keep failing, and warns you when a part is running low.
>
> I'll start with who uses the records and the words they use. Your ten technicians write the records, and your service manager and director read them. In a firm your size, the service manager is usually dispatcher, stock controller and analyst at once. So the reports are only as good as what technicians write, and published studies of real work orders find key fields are often missing or wrong.
>
> That changed how our loader works. It checks every row before it reaches the database. If the serial number is missing, the date is ambiguous, or the part number isn't in your P-nnnn format, it rejects the row and tells you which line and why, so it can be fixed at source."

(About 150 words.)

Likely questions and model answers

"Where in your sources did you learn how technicians describe faults and parts?"

This is where the sources are weakest. I found no study of the words IT field engineers use. For the formal meaning of fault and failure, I used the IEC's free vocabulary, Electropedia. For how technicians actually write, NIST researchers describe maintenance text as full of jargon and abbreviations, but their data is industrial, not IT. For UK words, a UK job advert uses "job sheets" and "stock on your vehicle". That is why my first question for you is to see a real job sheet.

“What should happen if a technician logs a part number in an old or different format?”

If the number is in the wrong shape, it is rejected with the line number, because we cannot safely guess which part it means. If it is in the right shape but not in your stock list, it is stored and flagged as an unknown part, so the visit is not lost. Real parts systems use a cross-reference table that maps old and manufacturer numbers to your number. We have not built that yet; it is the natural next step.

“Why reject a row instead of fixing it?”

We fix changes that need no guessing: extra spaces are removed and letters uppercased, so “p-1180” becomes “P-1180”. Anything that needs a guess is rejected. A date like 03/04/2026 could be March or April, and a missing serial number cannot be invented. The UK Government Data Quality Framework recommends fixing problems at source, and the tool reports each rejected row with its line number and reason, so the office can correct the sheet and load it again.

“Why is ‘failed’ an outcome, and is the list complete?”

The list came from the client brief, but my research suggests it needs work. “Failure” in the standard vocabulary is something equipment does, so `failed` could be misread. `replaced_part` is an action rather than a result. “No fault found” and “awaiting parts” are missing, so technicians will log both as `failed` and the manager cannot tell a hard fault from a missing part.

Source cautions

All quotes were read through web extracts. Open each source and confirm the exact wording before submitting.

- **Microsoft Learn, Fieldmotion, PTC:** vendor sources, used only to show common practice.
- **DWP advert:** an employer advert (furniture maker, not IT). Adverts expire, so save a copy.
- **Hodkiewicz & Ho, Khan et al.:** abstracts only; the full papers are paywalled.
- **Electropedia:** “inability to perform” in the brief is shortened from “inability to perform as required, due to an internal state”.
- **NATO:** the duplicate-prevention point is a summary of the fact sheet, not a direct quote.
- **Volvo:** documentation for one product, not a general standard.
- **Sexton & Brundage:** “mispellings” is the source’s own spelling; the data is industrial, not IT.

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Julian: How records are kept

Researched and written by **Julian Marchington**

Full service history protects evidence, recalls and stock

Field service companies keep a full, linked history of every visit and every part fitted for three reasons. Without it they cannot spot repeat faults or work out reliability figures. They cannot find the devices affected by a supplier recall. And they cannot prove to a customer, auditor or court what was done, when, and by whom. **No single law sets a 10-year retention period.** UK tax rules require about 6 years, contract claims can be brought for 6 years after the work, negligence claims have a 15-year longstop, and published UK retention schedules keep equipment maintenance records for “life of equipment + 6 years” or 10 years from acquisition. The client’s “10 years” therefore sits inside normal, defensible practice. On access, the quality standard ISO 9001 separates permission to view records from permission to change them, and asks that evidence records be “protected from unintended alterations”. Our tool fits this picture well: it saves each visit, its parts and its stock change in one transaction, skips duplicate visit IDs, never edits or deletes, and stores unknown parts instead of rejecting the visit. Three gaps need to be stated honestly: no logins or roles, no recorded way to correct a mistake, and no retention rule for the personal data (technician and customer contact names) inside the history.

How to read this report. The research was done on 1 November 2026 through a web-fetch tool that returns short extracts, so no quote has been checked character by character against the original page. Neutral sources (legislation, regulators, academic papers, a manufacturer recall notice, official software documentation) are preferred, and vendor sources are labelled. The legal sources are UK. This is general information for a university project, not legal advice.

A device’s history only becomes useful once every visit is linked to it

The strongest neutral evidence for a full history comes from **parts traceability**. Zebra Technologies’ voluntary recall of power supplies for its ZP5XX label printers covered “PSUs manufactured by the FSP Group between October 1, 2006 and December 31, 2012”, and said “the issue is limited to an identified batch of PSUs”. A printer was excluded if its power supply had already been replaced (Zebra Technologies [1]). Whether a customer’s printer was affected therefore depended on its **parts-replacement history**. A service company can only answer “which of our customers’ devices are affected?” if it recorded which part went into which device, and when.

The second reason is **finding repeat faults and measuring reliability**. A repeat failure only shows up when visits are linked to the same device over time; seen one at a time, each visit looks like a new fault. NIST researchers studying historical maintenance work orders found that “data quality often impacts an analyst’s ability to calculate important Key Performance Indicators (KPIs)” (Conte et al., PHM Society 2021 [2]). The standard measures are simple sums over the visit history: **MTBF = Total Operating Time ÷ Number of Failures** (ReliaMag [3]). The arithmetic makes the point without any vendor claim. A machine that runs for 12 months and fails 3 times fails on average every 4 months. Lose two of those visit records and it appears to fail once every 12 months, so **equipment looks three times more reliable than it is** (an illustration of the formula, not a study figure).

The third reason is **evidence and stock planning**. The visit record is what supports an invoice, a warranty claim or a dispute, and average parts usage, which drives reorder levels, comes straight from the history of parts used on visits. Hard industry numbers are scarce. The only benchmark found was Aberdeen Group's first-time fix rates (88% best-in-class, 80% average, 63% laggards), seen only second-hand through a vendor page (PTC [4], vendor), so it is treated as indicative only.

No law sets ten years, but several drivers land close to it

ISO 9001:2015 clause 7.5.3 asks organisations to address "retention and disposition" of documented information, but it **sets no period**; each firm must decide its own (Auditor Training Online [5], secondary source quoting the paywalled standard). The actual period comes from other drivers.

Driver	Period	Source
UK company tax records	6 years from end of the financial year	GOV.UK [6]
UK VAT records	"at least 6 years"	HMRC VAT Notice 700/21 [7]
Simple contract claims (England & Wales)	6 years from the cause of action	Limitation Act 1980 s.5 [8]
Negligence, latent damage (e.g. a faulty repair causing a later fire)	15-year longstop from the act	Limitation Act 1980 s.14B [8]
UK Government Mid-Tier contracts	"seven (7) years after the End Date" of the contract	Cabinet Office [9]
NHS England equipment maintenance records	10 years from acquisition	NHS England, v4.0 [10]
Heriot-Watt University equipment maintenance logs	"Life of equipment + 6 years"	Heriot-Watt [11]

Tax law alone stops at about 6 years, so **the client's 10 years has to rest on other drivers**. A 6-year contract claim runs from each visit and a 15-year negligence longstop runs from the act, so a flat 10 years sits inside the risk window. Contracts add more: a 5-year maintenance contract followed by a 7-year post-contract records clause means records of early visits may be needed about 12 years after they were made. Published schedules follow the pattern **life of the asset plus 6 years**; for servers and switches with a 5 to 10 year life, that gives roughly 11 to 16 years, which supports the client's figure. These periods are policy choices, not fixed law.

Records should be viewable by several people, changeable by few, and never silently overwritten

ISO 9001:2015 clause 7.5.3 asks firms to control "distribution, access, retrieval, and use" of records. A note to the clause splits access into "the permission to view the documented information only" or "the permission and authority to view and change the documented information", and the clause ends: "Documented information retained as evidence of conformity shall be protected from unintended alterations" (Auditor Training Online [5], secondary). The split maps directly onto the client's roles: ten technicians create visit records, and the service manager and director view the history.

Regulated industries spell out how changes should work. The UK medicines regulator (MHRA) says an audit trail should show "the 'who, what, when and why'" of a change, recorded "without obscuring or overwriting the original record", and that effort "should be commensurate with the

risk” (MHRA GxP Data Integrity Guidance, 2018 [12]). **This does not bind a field service company**; it is used as the clearest published model of good practice. MHRA also says “shared logins or generic user access should not be used”, because without individual identities the “who” in the record means nothing.

Personal data inside the history runs on its own, shorter clock

Most of the history is about equipment and is generally not personal data. But each visit names a technician and often a customer contact, and that data falls under UK GDPR, which says personal data must be “kept in a form which permits identification of data subjects for no longer than is necessary” (UK GDPR Art.5 [13]). The ICO confirms “The UK GDPR does not set specific time limits for different types of data” and “It is up to you to justify this”; when data is no longer needed, “You can either erase (delete) it, or anonymise it” (ICO storage limitation [14]). So keeping history indefinitely is defensible for the equipment record but **not, by default, for the names in it**.

The tool’s design choices mostly match good practice, with three honest gaps

Design choice	What supports it	Gap or next step
One SQLite transaction per visit (visit, parts, stock change)	SQLite: “either all database changes within a single transaction occur or none of them occur”, even after “an operating system crash or power failure” (SQLite atomic commit [15])	Roll back the whole visit on any error, and keep SQLite’s journal switched on
Stock may go negative	Recording what was used matters more than blocking the visit	A negative level needs someone to reconcile it, through a recorded adjustment rather than an overwrite
Unknown part numbers stored and flagged (no foreign key to stock)	Keeps the visit record complete	Someone must review flagged parts; they do not feed reorder figures until matched
Duplicate visit IDs skipped, not overwritten	Changes should not obscure the original record (MHRA, by analogy)	Report skipped IDs, so a genuinely different visit with a reused ID is noticed
No edit or delete	ISO 9001: evidence “protected from unintended alterations”	**No correction mechanism.** Next step: a correction record pointing to the original visit, with author, time and reason
Who reads the history	Client said service manager and director; ISO 9001 view-only vs view-and-change split	**No logins or roles.** Access is limited only by who can reach the server and database file
History kept indefinitely	Client said 10 years; limitation periods and UK schedules support 10+ years	**No retention rule for personal fields.** Next step: written retention schedule, then anonymise names after it
Docker volume, backups checked with `PRAGMA integrity_check`	Backup is copied with SQLite’s backup API and checked before reporting OK	One copy on one host. Add `PRAGMA foreign_key_check`, off-host copies and regular test restores

Conclusion

“10 years” is not a legal number; it is a sensible policy that the team can justify with limitation periods and published schedules, and it is best measured from the end of a contract or the retirement of a device, not from the date of each record. “Accessed by service manager and

director” describes reading, but the harder question is who may *correct* a record, and today’s tool answers “nobody”. That is safe against silent tampering, but it is not the same as good change control. The most useful single upgrade is a correction record with author, time and reason. Logins, a written retention schedule with anonymisation of personal fields, and off-host backups with test restores come next. Stating these as planned work, rather than claiming compliance, is both more honest and stronger in front of a client.

Draft client brief section: How records are kept

Researched and written by Julian Marchington

Why a full history. A repeat fault only shows up when every visit is linked to the same device. Researchers at NIST found that poor work-order data “often impacts an analyst’s ability to calculate important Key Performance Indicators” [2]. Recalls depend on parts history too: Zebra’s printer power-supply recall covered “an identified batch of PSUs”, and printers whose power supply had already been replaced were excluded [1]. Only a record of which part went into which device can answer that.

How long. No single law says 10 years. ISO 9001 asks each firm to set its own “retention and disposition” rules [5]. In the UK, tax records are kept 6 years [6], contract claims can be brought for 6 years after the work, and negligence claims have a 15-year longstop [8]. NHS England keeps equipment maintenance records for 10 years from acquisition [10]. Your 10 years fits this practice.

Who may see or change records. ISO 9001 separates permission “to view the documented information only” from permission “to view and change”, and says evidence records “shall be protected from unintended alterations” [5]. In regulated industries, changes are recorded “without obscuring or overwriting the original record” [12]; we use this as a model, not as a legal duty for you.

Personal data. Technician and customer contact names fall under UK GDPR, which allows keeping them “no longer than is necessary”; the ICO says firms set and justify their own periods, then erase or anonymise [13, 14].

What this means for the tool. Each visit, its parts and its stock change are saved in one SQLite transaction, so “either all database changes within a single transaction occur or none of them occur” [15]. Records are never edited or deleted, and duplicate visit IDs are skipped rather than overwritten. Stock may go negative and unknown parts are kept and flagged, so no visit is lost. Not yet built: logins, a recorded way to correct mistakes, and a retention rule for personal fields. Backups are checked with `PRAGMA integrity_check`; we plan to add off-host copies and regular test restores.

The numbers in this section point to the References at the end of Julian’s research.

Assumptions log rows

Decision	What we assumed	Source or reason	Who decided
One transaction per visit	Visit, parts used and stock change are saved together or not at all; on any error the whole visit is rolled back.	SQLite atomic commit. A half-saved visit (parts without a visit, or stock moved with no reason) is worse than a failed save the technician can retry.	Julian
Stock may go negative	A technician's record of what was fitted is the fact; the stock count may be wrong. Negative stock is flagged as a shortfall.	Completeness of the record comes first. **Diverges from typical stock control**, where stock cannot fall below zero. Future work: a recorded stock adjustment with reason.	Julian
Unknown parts stored and flagged (no foreign key to stock)	New or old part numbers will appear that are not in the stock list; storing and flagging keeps the visit.	Same completeness reason. Unknown parts are not counted in reorder figures until matched. Also meant release 1.1.0 needed no schema change.	Julian
Duplicate visit IDs skipped, not overwritten	A repeated visit ID is most likely a re-import, so the first stored record is kept unchanged.	Changes should not obscure the original record (MHRA, by analogy). Risk: a genuinely different visit reusing an ID is dropped; future work: report and compare skipped IDs.	Julian
History kept, no delete feature (client said 10 years)	Keeping everything meets the client's 10 years and protects the evidence chain.	Client meeting; Limitation Act 1980; NHS England schedule. **Diverges from best practice:** ISO 9001 expects a defined retention rule, and UK GDPR storage limitation applies to names. No correction mechanism yet. We do not claim GDPR compliance today.	Julian
Who can read the history	The service manager and director read all history; technicians create visit records.	Client meeting; ISO 9001 view-only vs view-and-change access. **No logins or roles exist yet.** MHRA advises against shared logins. Future work: individual logins and roles.	Julian

Questions for the client

1. You said records are needed for 10 years. Should that run from the date of each visit, from the end of a customer's contract, or from when a device is retired? Do any of your customer contracts require you to keep records for a set time after the contract ends?
2. When a technician makes a mistake on today's paper sheet, how is it corrected, and who is allowed to make that correction?
3. Do your paper sheets record the serial or batch number of parts fitted, or only the part number? Have you ever had to find every device that received a part from a recalled batch?
4. When your stock count and what technicians say they used do not match, what is the usual cause, and who sorts it out today?
5. Do the service manager and director need to see technician and customer contact names in old history, or mainly the equipment, faults and parts?
6. Will the existing paper records be typed into the tool? If so, could two sheets share the same visit number?

One-minute presentation opener

> "My research looked at how service companies keep records, and it gave us three requirements.

> First, the history has to be complete and linked. SW-4410's three visits in twenty-two days only show up when every visit is tied to the same serial number. It keeps reliability figures honest too: if a machine runs for twelve months and fails three times, that's one failure every four months. Lose two of those visit records and it looks like one every twelve months. And when Zebra recalled printer power supplies built over six years, only a parts history could tell a service company which customers were affected.
>
> Second, retention. No single UK law sets ten years. Tax and VAT records need six, contract claims run six years after the work, and negligence has a fifteen-year longstop. NHS England keeps equipment maintenance records for ten years. Ten years is a defensible policy.
>
> Third, integrity. ISO 9001 separates permission to view records from permission to change them. So our rule is: records are added, never silently overwritten."

(About 175 words.)

Likely questions and model answers

"What did your research say about why the company needs a full history?"

Four reasons. You can only spot a repeat fault when every visit is linked to the same machine. Reliability figures are sums over the history, so deleting old visits makes equipment look better than it is. When a supplier recalls a batch of parts, only a parts history tells you which customers are affected, as Zebra's printer recall showed. And the record is the evidence for an invoice, a warranty claim or a dispute. If pushed for an industry figure: Aberdeen Group puts first-time fix rates at 88% for the best firms, 80% on average and 63% for the weakest, but that was seen second-hand through a vendor site, so treat it as indicative only.

"Why save a visit, its parts and the stock change in one transaction?"

Because they describe one real event, and they must all be stored or none of them. SQLite guarantees a transaction commits completely or not at all, even after a crash or power cut. Without it, a failure halfway could reduce stock with no visit to explain it, which is exactly what loses a billing dispute.

"Why ten years and not six or forever?"

Six only covers tax. Contract claims run six years from each visit, negligence up to fifteen, and a five-year maintenance contract with a seven-year records clause means early visits may be needed about twelve years later. Forever isn't right for the names in the records, because UK GDPR says personal data can't be kept longer than necessary. So ten years for the equipment history, with a separate, shorter rule for names, is the honest answer.

"What can't your tool do yet?"

Three things. There are no logins, so anyone who can reach the file can read it. There's no recorded way to correct a mistake. And there's no rule yet for removing technician and contact names after a set time. The most useful next step is a correction record that keeps the original and notes who changed it, when and why.

Source cautions

All quotes were read through short web extracts. Open each source and confirm the exact wording before submitting.

- **ISO 9001 (Auditor Training Online):** the standard is paywalled; its wording was seen only in this secondary source.
- **Zebra:** the “already replaced units excluded” point is a paraphrase; the page is a Latin America (/la/es/) URL.
- **NHS England:** the schedule was read through a text extractor and the table may be misaligned; confirm row 5.8.
- **MHRA:** pharmaceutical guidance, used as a model of good practice only; it does not bind the client.
- **PTC (Aberdeen figures):** second-hand; the original Aberdeen report was not opened.
- **ICO:** possibly a legacy URL; find the current page and confirm the wording.
- **Heriot-Watt:** schedule dated 2010; check it is still current.

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